the same utility for all the portfolios on a given indifference curve. The investor moves to higher utility by moving to successively higher indifference curves. The right graph in the top row is the mean-variance frontier that we constructed from section 2. The frontier is a property of the asset universe, while the indifference curves are functions of the risk aversion of the investor.

We bring the indifference curves and the frontier together in the bottom row of Figure 3.8. We need to find the tangency point between the highest possible indifference curve and the mean-variance frontier. This is marked with the *X*. Indifference curves lying above this point represent higher utilities, but these are not attainable—we must lie on the frontier. Indifference curves lying below the *X* represent portfolios that are attainable as they intersect the frontier. We can, however, improve our utility by shifting to a higher indifference curve. The highest possible utility achievable is the tangency point *X* of the highest possible indifference curve and the frontier.

Let's go back to our G5 countries and take a mean-variance investor with a risk aversion coefficient of $\gamma = 3$. In Figure 3.9, I plot the constrained (no shorting) and unconstrained mean-variance frontiers constructed using U.S., Japanese, U.K., German, and French equities. The indifference curve corresponding to the maximum achievable utility is drawn and is tangent to the frontiers at the asterisk. At this point, both the constrained and unconstrained frontiers overlap. The optimal portfolio at the tangency point is given by:

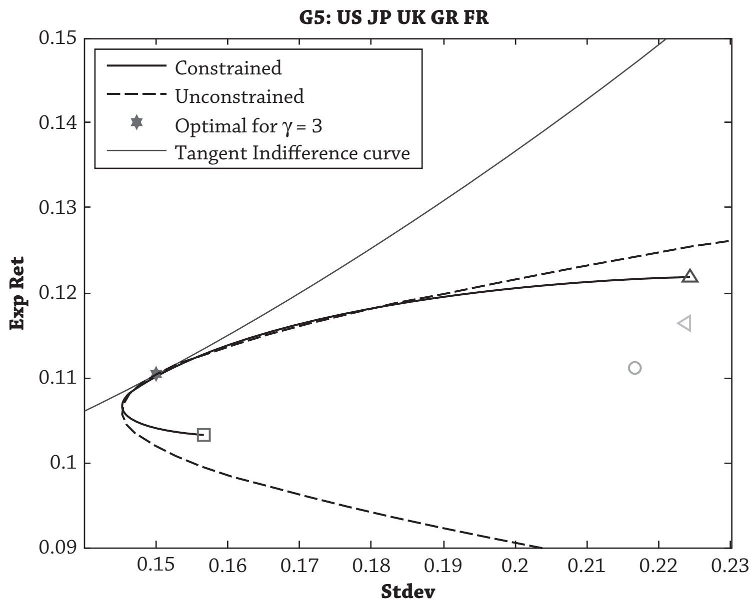

Figure 3.9